SRS — ms_05 Distribution

1. Fungsi

Mengelola pengiriman barang dari **warehouse ke retail** untuk memenuhi restock request: pembuatan shipment, penugasan kurir, pelacakan status, dan pemberitahuan kedatangan.

2. Kebutuhan

- Shipment dibuat otomatis dari event `shipment.requested` (idempoten per `restock_request_id`).
- Status: `CREATED → SHIPPED → DELIVERED` (atau `FAILED`); setiap perubahan dicatat di `shipment_trackings`.
- Kurir memperbarui status; admin/manajer warehouse dapat menugaskan kurir.
- Penerbitan event setiap perubahan status penting.

3. Endpoint

Method	Path	Deskripsi	Peran
GET	`/shipments`	Daftar (filter status, retail, warehouse)	WAREHOUSE_MANAGER, RETAIL_MANAGER, OWNER
GET	`/shipments/{id}`	Detail + tracking	idem + COURIER
PATCH	`/shipments/{id}/assign`	Tetapkan kurir	WAREHOUSE_MANAGER
PATCH	`/shipments/{id}/status`	Ubah status `{status, note}`	COURIER
GET	`/couriers/me/shipments`	Tugas kurir	COURIER

4. Model Data

`shipments`, `shipment_items`, `shipment_trackings` — lihat er.puml.

5. Event

- Consume: `shipment.requested`.
- Publish: `shipment.created`, `shipment.shipped`, `shipment.delivered`, `shipment.failed`.

6. Diagram

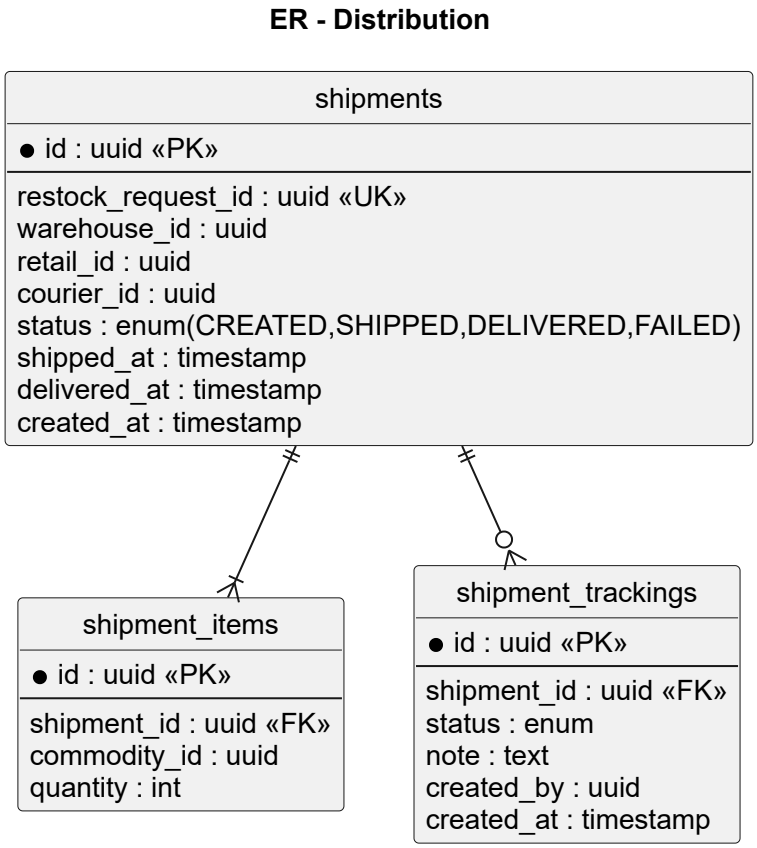
usecase.puml · er.puml · sequence_shipment.puml · state_shipment.puml

7. Cara Pembuatan

1. Database `db_distribution`; migrasi tiga tabel + `processed_events`, `outbox_events`; unik `restock_request_id`.
2. Consumer `shipment.requested` → buat shipment + tracking awal dalam satu transaksi.
3. Mesin status dengan validasi transisi; publisher event via outbox.
4. Handler REST dan pembatasan akses kurir hanya pada shipment miliknya.
5. Uji: event ganda, transisi ilegal, otorisasi kurir.

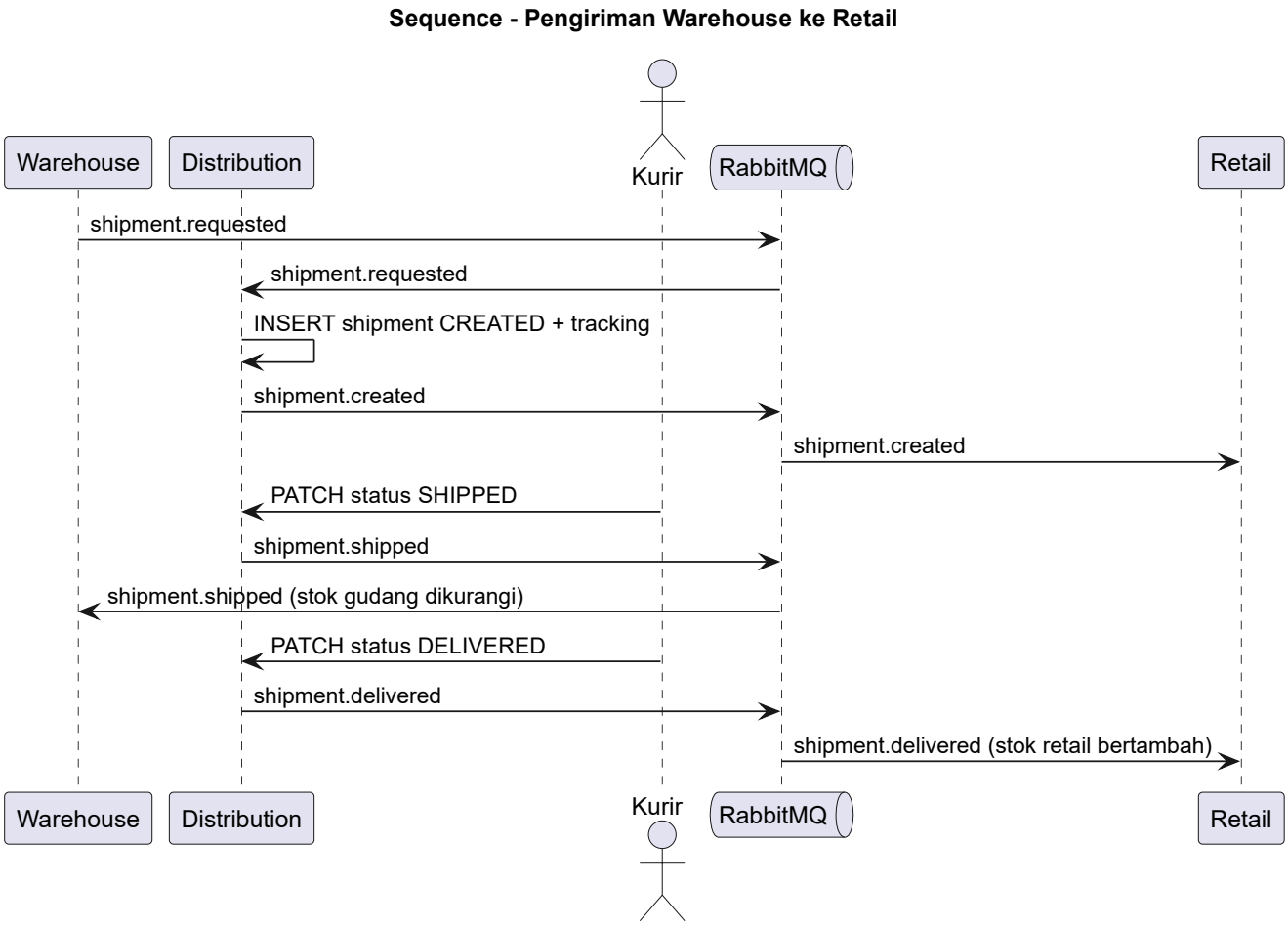
Lampiran Gambar Diagram

ER - Distribution



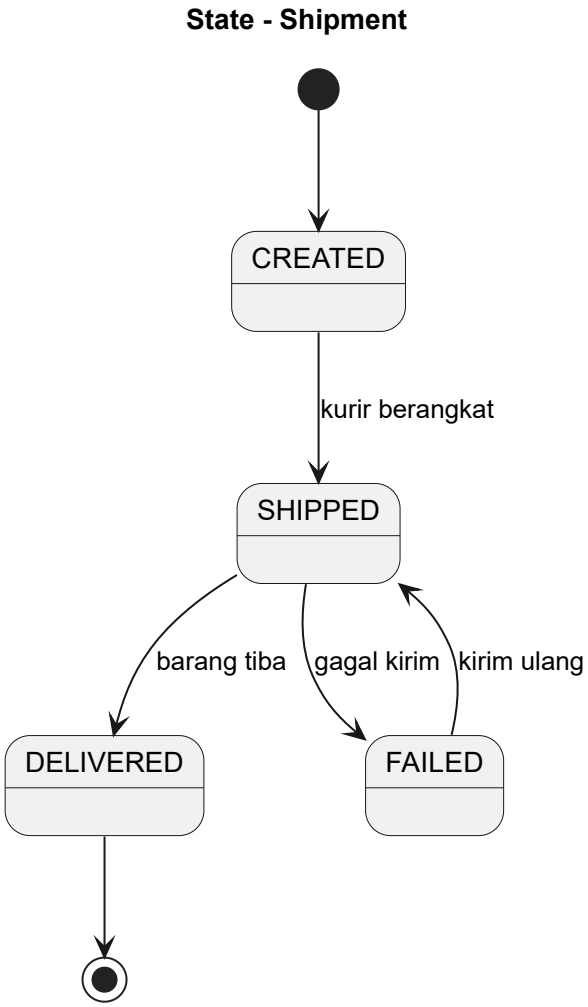
Sumber: er.puml

Sequence - Pengiriman Warehouse ke Retail



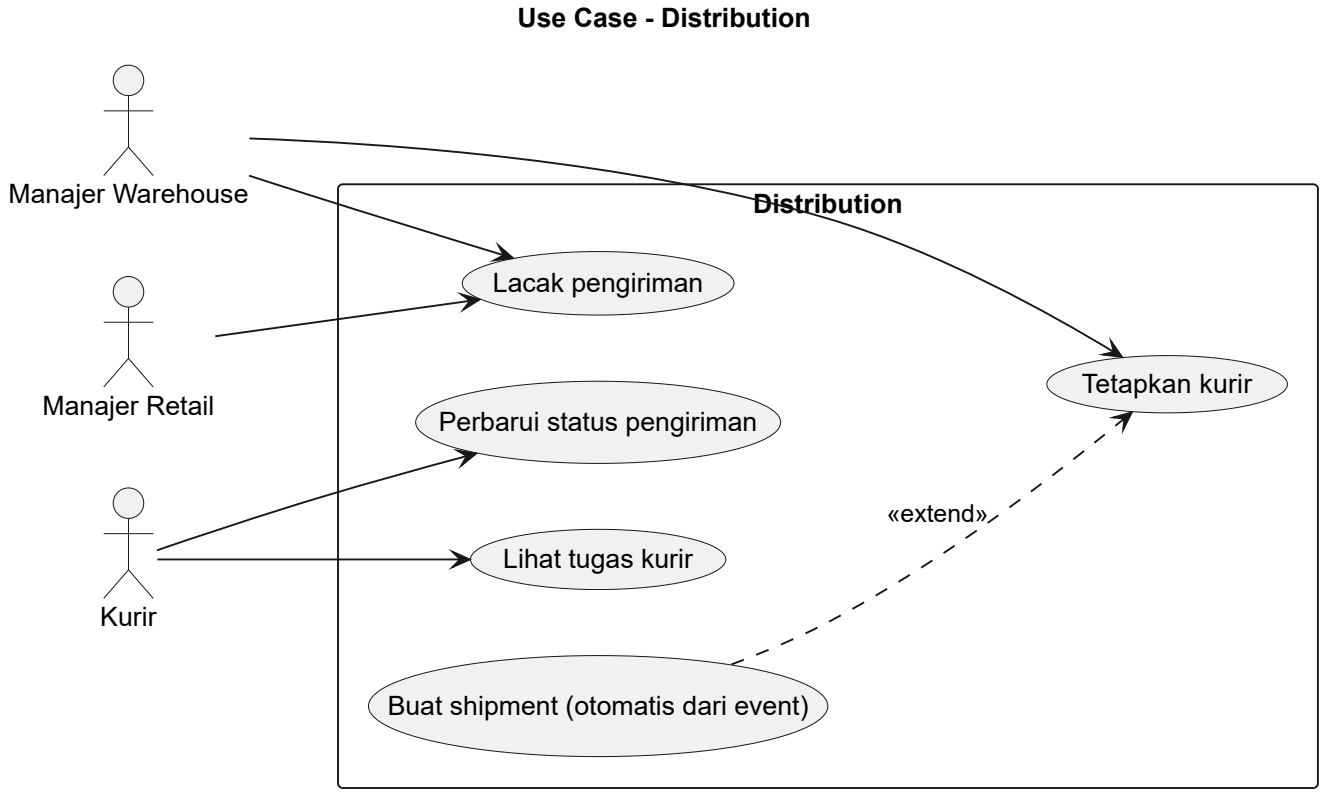
Sumber: sequence_shipment.puml

State - Shipment



Sumber: state_shipment.puml

Use Case - Distribution



Sumber: usecase.puml